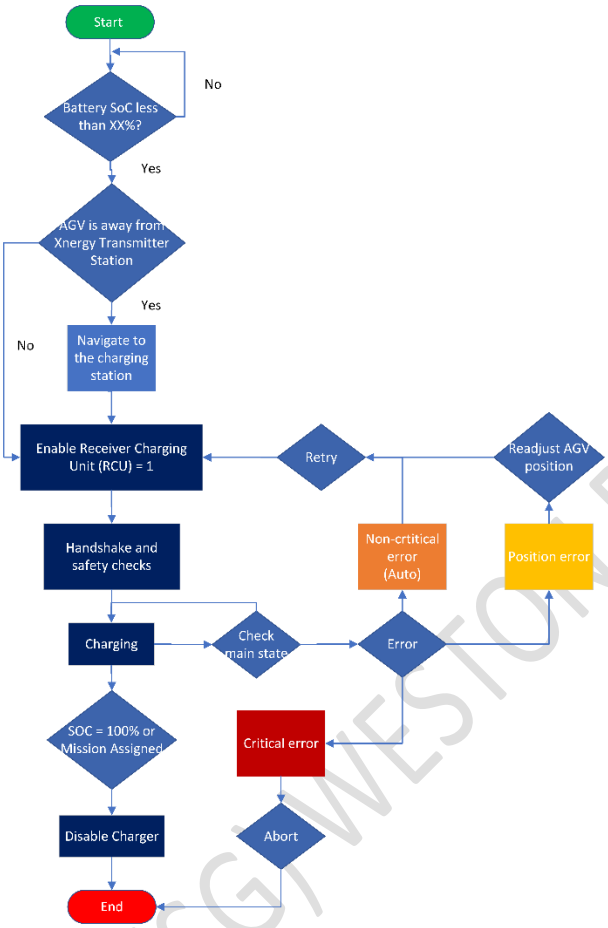
Basic Flowchart on Controlling Xnergy RCU



Position error: 0x40, 0x80, 0x10000
Non-critical error: 0x1, 0x2, 0x4, 0x8, 0x10, 0x20, 0x100, 0x200, 0x400, 0x800, 0x1000
Critical error: 0x2000, 0x4000, 0x8000, 0x20000

The basic flowchart, as illustrated above, is to allow users to control Xnergy Receiver Charging Unit (RCU) before wireless charging operation begins or when an error is alarmed during the wireless charging operation.

It is recommended to have the robot monitor both the battery data (SoC, cell voltage, cell temperature and warning error) and Xnergy RCU.

If the robot detects the battery below a certain SoC threshold (for example, 88%) and if the robot does not have any mission, robots may travel back to Xnergy Transmitter Charging Station. Once the robot has parked at the charging station, the robot may send a charging command to RCU, and Xnergy RCU will start to perform handshake and safety checks. Once successful, wireless charging operation begins.

If errors are detected during handshake or charging, errors will indicate the issues encountered. For example, 0x40 and 0x80 mean RCU is too far or too near Transmitter Pad; it is recommended for the robot to adjust its position again. For other errors, the robot may retry or even stop and abort wireless charging to inform Xnergy.

Once the robot detects SoC equals 100% or receive any new mission, it is highly recommended that the robot send a disable charging command to stop wireless charging operation before moving off.

ORIGIN FOR ISG WESTON ROBOT

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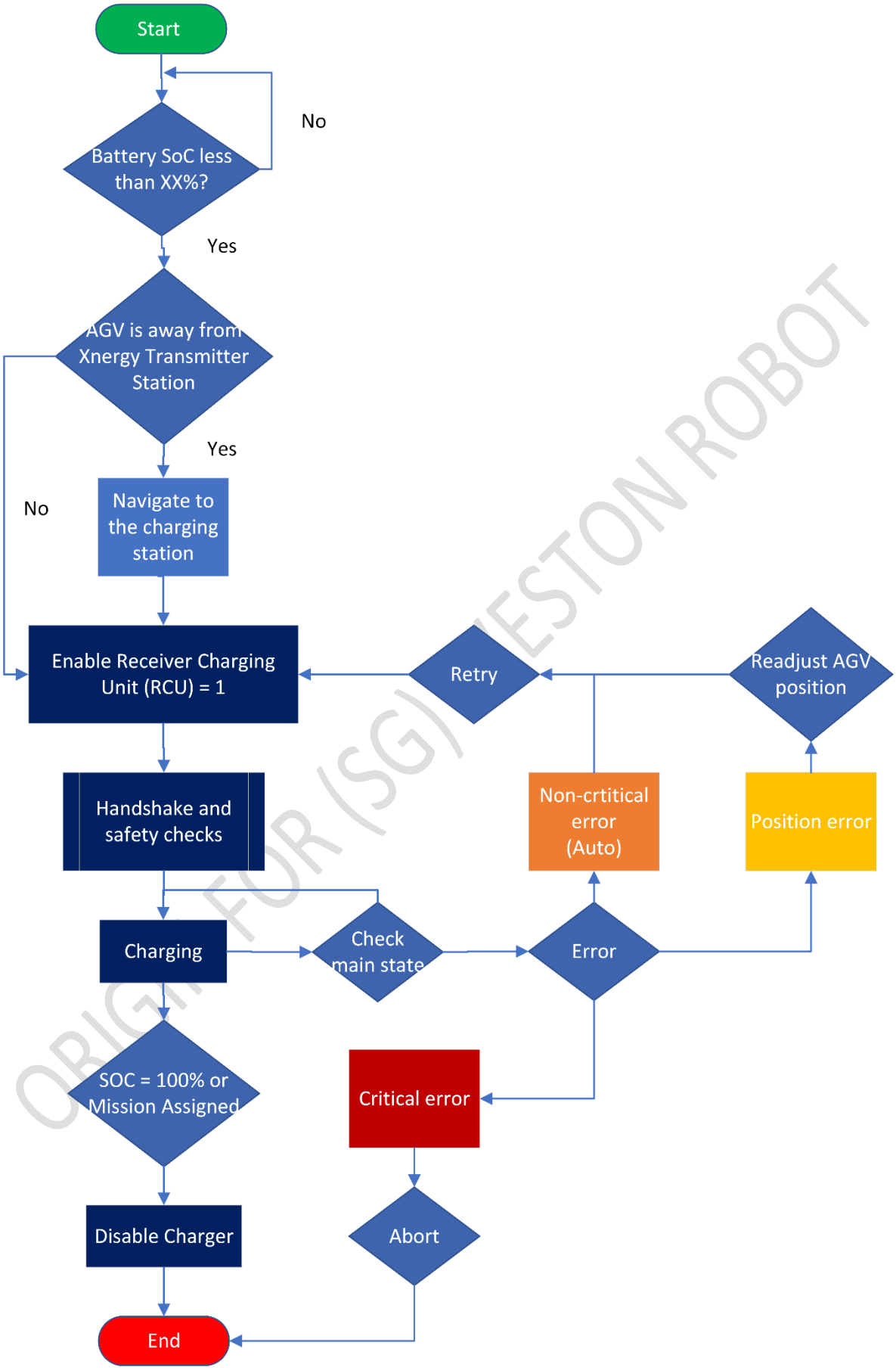

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